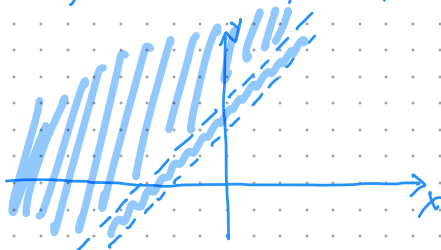


Question 1. Sketch the domain of $f(x, y)$.

(1) $f(x, y) = \frac{1}{\ln(y-x-2)}$

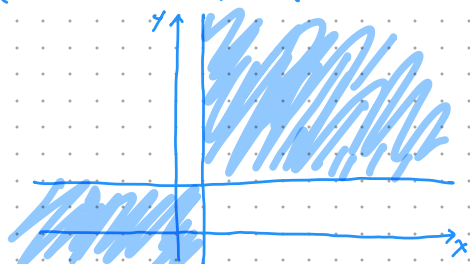
$\ln(y-x-2) \neq 0$
 $y-x-2 > 0$ and $y-x-2 \neq 1$



(3) $f(x, y) = \sqrt{(x-1)(y-2)}$

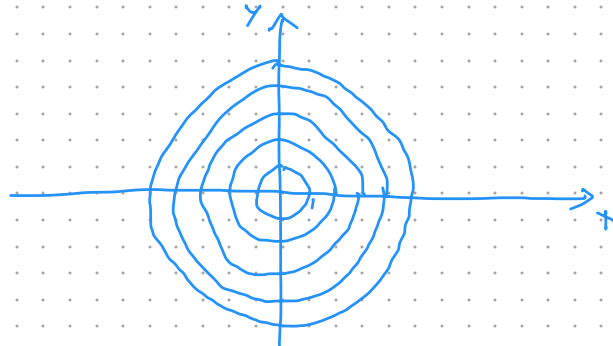
$(x-1)(y-2) \geq 0$

$(x-1 \leq 0 \text{ and } y-2 \leq 0) \text{ or } (x-1 \geq 0 \text{ and } y-2 \geq 0)$

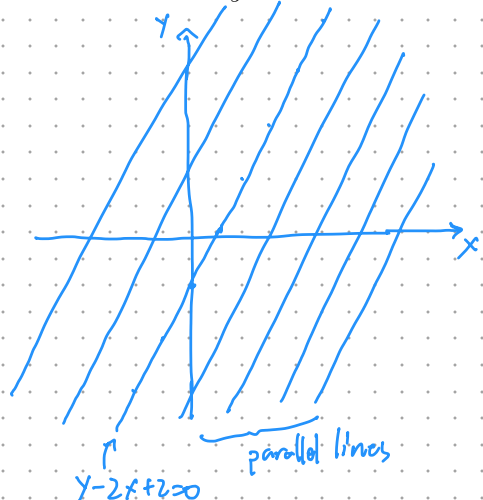


Question 2. Sketch the level curves of $f(x, y)$.

(1) $f(x, y) = \sqrt{x^2 + y^2}$

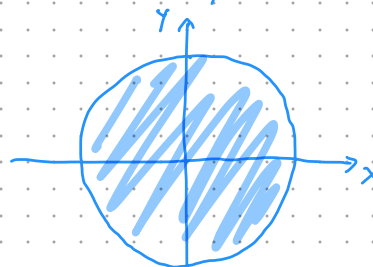


(3) $f(x, y)$ is the distance of the point (x, y) from the line $y - 2x + 2 = 0$.



(2) $f(x, y) = \sqrt{4 - x^2 - y^2}$

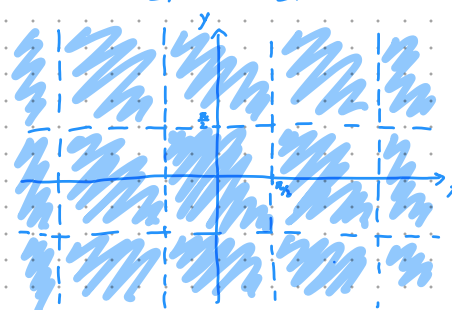
$4 - x^2 - y^2 \geq 0$
 $x^2 + y^2 \leq 4$



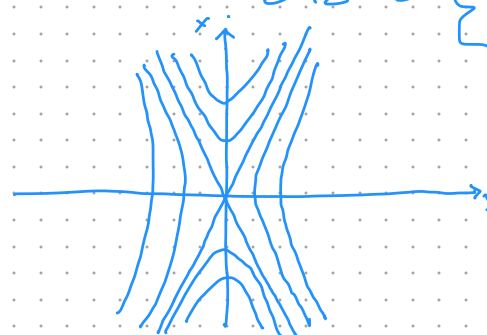
(4) $f(x, y) = \sec(x) \tan(y)$

$x \neq \dots, -\frac{3\pi}{2}, -\frac{\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{2}, \dots$

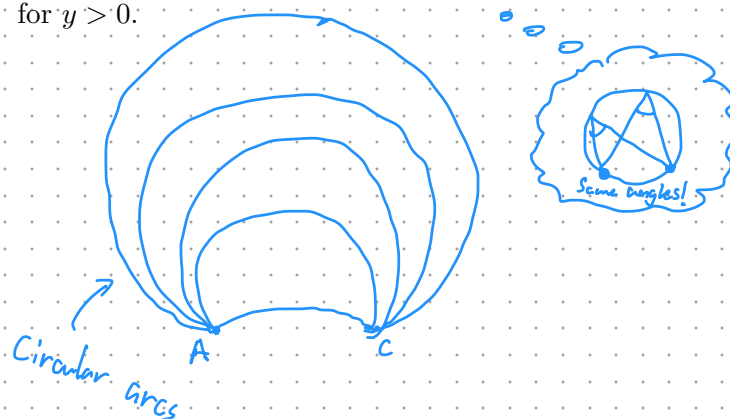
$y \neq \dots, -\frac{3\pi}{2}, -\frac{\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{2}, \dots$



(2) $f(x, y) = 4x^2 - y^2$



(4) (challenge) $f(x, y)$ is the angle ABC formed by $A = (-1, 0)$, $B = (x, y)$, and $C = (1, 0)$ for $y > 0$.



Question 3. Evaluate the following limits or show that the limit does not exist.

$$(1) \lim_{(x,y) \rightarrow (0,\pi/4)} \sec(x) \tan(y)$$

↑
Continuous at $(0, \frac{\pi}{4})$,
Just plug it in

$$= \sec(0) \tan\left(\frac{\pi}{4}\right)$$

$$= 1$$

$$(2) \lim_{\substack{(x,y) \rightarrow (1,1) \\ x \neq y}} \frac{x+y-2}{x-y}$$



① Along $y=1, x \rightarrow 1$

$$\lim_{x \rightarrow 1} \frac{x+1-2}{x-1} = \lim_{x \rightarrow 1} \frac{x-1}{x-1} = 1$$

② Along $x=1, y \rightarrow 1$

$$\lim_{y \rightarrow 1} \frac{1+y-2}{1-y} = \lim_{y \rightarrow 1} \frac{y-1}{1-y} = -1$$

Limits disagree along two paths, so the original limit DNE.

$$(3) \lim_{(x,y) \rightarrow (1,1)} \frac{x-y}{x^2-y^2}$$

$$\stackrel{x \neq y}{=} \lim_{(x,y) \rightarrow (1,1)} \frac{x-y}{(x-y)(x+y)}$$

$$= \lim_{(x,y) \rightarrow (1,1)} \frac{1}{x+y}$$

$$= \frac{1}{2}$$

$$(4) \lim_{(x,y) \rightarrow (0,0)} \frac{2xy}{x^2+y^2}$$

Consider the path $y=mx$

$$\lim_{x \rightarrow 0} \frac{2x(mx)}{x^2+(mx)^2} = \frac{2m}{1+m^2}$$

We can find different m 's such that the limit differs, so the original limit DNE.

$$(5) \lim_{(x,y) \rightarrow (0,0)} x \cos \frac{1}{y}$$

$$\stackrel{y \neq 0}{-|x|} < x \cos \frac{1}{y} < |x|$$

$\lim_{(x,y) \rightarrow (0,0)} (-|x|) = \lim_{(x,y) \rightarrow (0,0)} |x| = 0$, so by sandwich theorem, the original limit is 0.

$$(6) \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^2 + y^2}$$

Use $x = r \cos \theta, y = r \sin \theta$

$$= \lim_{r \rightarrow 0^+} \frac{(r \cos \theta)^2 (r \sin \theta)}{r^2}$$

$$= \lim_{r \rightarrow 0^+} r \cos^2 \theta \sin \theta$$

$$0 \leq r \cos^2 \theta \sin \theta \leq r \quad (\text{for positive } r)$$

So by sandwich theorem, the limit is 0.

$$(7) \lim_{(x,y) \rightarrow (0,0)} \frac{\sin(x^2 + 2y^2)}{x^2 + y^2}$$

Along $y=0, x \rightarrow 1$

$$\lim_{x \rightarrow 1} \frac{\sin(x^2)}{x^2} = 1$$

Along $x=0, y \rightarrow 1$

$$\lim_{y \rightarrow 1} \frac{\sin(2y^2)}{y^2} = 2$$

The limits along different paths differ, so the original limit DNE.

$$(8) \lim_{(x,y) \rightarrow (0,0)} e^y \frac{\sin x}{x}$$

$\stackrel{x \neq 0}{\text{Note that } f(x) = \begin{cases} \frac{\sin x}{x} & x \neq 0 \\ 1 & x = 0 \end{cases}}$ is continuous at $x=0$,

$$= \lim_{(x,y) \rightarrow (0,0)} e^y f(x) = e^0 f(0) = 1$$